

Lumen: an Open, Differentiable, GPU-Parallel Environment for Endovascular AI

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Abstract

Endovascular robotics and embodied medical AI need simulation environments that expose the actual control problem: a slender instrument threading a deformable, branching, image-guided lumen under safety constraints. Lumen is an open Apache-2.0 simulator for that setting. It combines tube-intrinsic contact, a deformable anisotropic vessel wall, torsion, friction, flow/clot coupling, synthetic fluoroscopy, pixel labels, Gymnasium environments, and a safety-scored benchmark in one public stack. The simulator is built on NVIDIA Newton and Warp, so the same code path supports differentiable mechanics and GPU-parallel rollouts. Lumen is designed as a research substrate for reinforcement learning, computer vision, imitation learning, and dataset tooling: it can generate procedural anatomy, render paired schematic and fluoroscopic views, validate replayable case bundles, build dataloader indexes, split episodes without leakage, and materialize strict tensor batches. This preprint describes the design principles, implementation surface, benchmark semantics, and relationship to prior open endovascular simulation work such as CathSim. The public repository is <https://github.com/SeldingerMed/seldinger-lumen>; project documentation and demos are at <https://seldingermed.github.io/seldinger-lumen/>.

1 Motivation

Endovascular procedures are a natural frontier for embodied AI. The task is precise, spatially constrained, image-guided, and safety-critical: a device must be advanced and rotated through a vascular lumen while maintaining wall safety, selecting branches, and responding to contact, tortuosity, stenosis, clot, and flow. Robotic systems already make remote and radiation-sparing catheter operation realistic, while reinforcement learning and imitation learning offer a route to policies that can reason over image feedback and contact-rich instrument dynamics [3, 4, 5, 6].

The bottleneck is not merely the availability of a Gym API. The bottleneck is whether a simulator can make clinically meaningful failures visible to learning algorithms. A rigid tube with a target-reaching reward can produce an agent that reaches the goal by scraping through the wall. A pixel renderer without segmentation and keypoints creates vision data but not a complete CV training artifact. A mechanical solver without replayable episodes is hard to use as a dataset generator. Lumen is built around the opposite premise: the mechanics, labels, benchmark, and data artifacts should be part of the same environment.

2 Design goals

Lumen is organized around five design goals.

- 1. The lumen is a stateful physical object.** The wall is not just a collision mesh. It is represented as a shared lumen field with anisotropic deformability and contact response. This lets the same quantity define the contact boundary, wall displacement, and safety metrics.

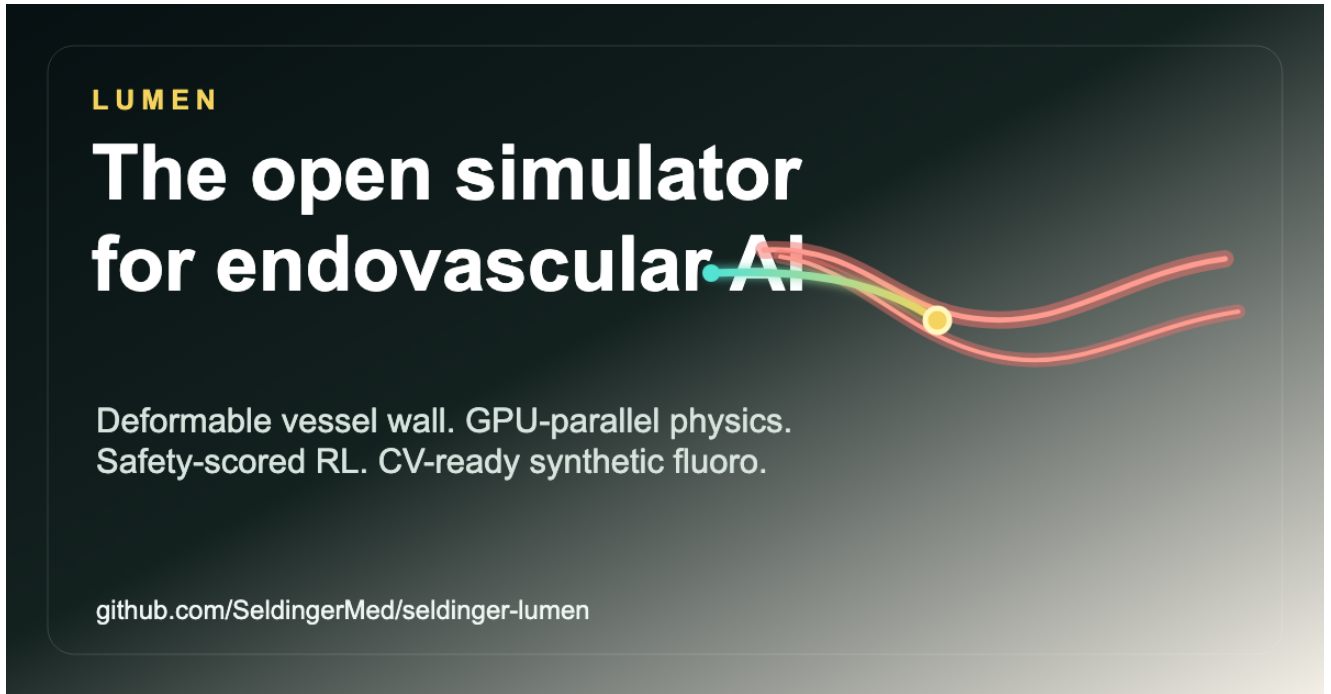


Figure 1: **Lumen launch overview.** Lumen exposes deformable-lumen physics, GPU-parallel simulation, safety-scored reinforcement learning, and CV-ready synthetic fluoroscopy as a single open research stack.

2. Contact should be tube-intrinsic. A centerline parameterization provides local coordinates for projecting the device into the vessel. Contact is expressed in terms of radial excess relative to the current lumen field and injected into the Newton/Warp solve. This avoids treating the vessel as an external triangle mesh when the control problem is fundamentally luminal.

3. RL success must be safety-scored. The benchmark reports raw target reach, safe success, unsafe success, maximum wall penetration, and return. The leaderboard ranks safe success before raw success, so wall-violating policies do not dominate by reward hacking.

4. Vision data should be first-class. Lumen can emit synthetic fluoroscopy and luminal observations with device masks, vessel masks, tip/base keypoints, node locations, actions, clinical metrics, and calibration metadata.

5. The artifact should be reusable. The open core is Apache-2.0, provides typed Python APIs, exposes CLI workflows, and writes replayable case bundles rather than one-off screenshots.

3 System overview

Figure 2 summarizes the stack. Procedural anatomy feeds a Newton/Warp solver with a deformable lumen field. The solver produces instrument states, wall-safety signals, and sensor inputs. Sensor models render schematic and fluoroscopic observations. Dataset tools validate, index, split, and materialize episodes for downstream RL and CV training.

3.1 Mechanics

The device is represented as a slender continuum instrument running inside a tubular or branching asset. Lumen uses a Newton/Warp backend for rod dynamics and differentiable simulation, following the broader trend toward programmable differentiable physics systems for robot learning [10, 12, 11, 14].

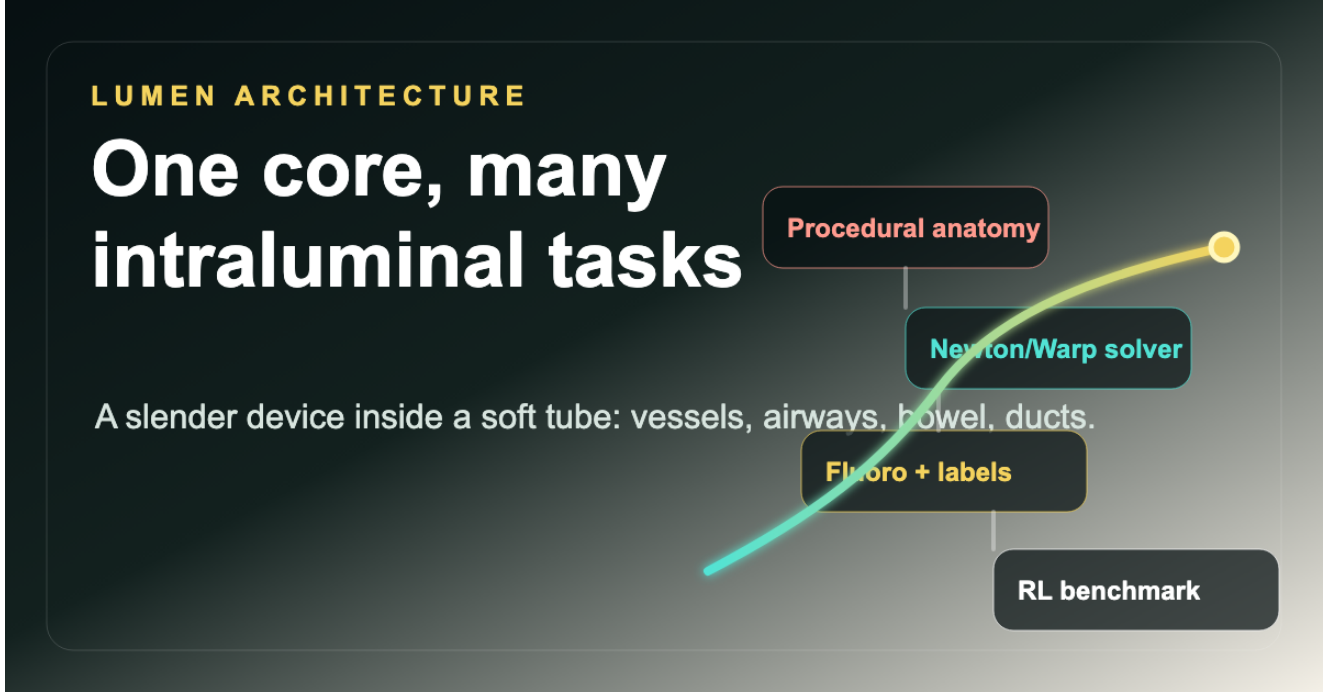


Figure 2: **Lumen architecture.** The same core supports procedural anatomy, differentiable mechanics, synthetic fluoroscopy, CV labels, and safety-scored RL benchmarks.

The wall model is inspired by anisotropic arterial mechanics, especially Holzapfel–Gasser–Ogden families of constitutive models for collagen-reinforced arterial tissue [19, 20]. The rod/contact design draws on elastic rod simulation and constrained time integration work in graphics and robotics [15, 16, 17, 18].

3.2 Observations

Lumen exposes compact state observations for fast control experiments and image observations for CV/RL pipelines. Synthetic fluoroscopy includes paired image and label products: device/vessel masks, device node positions, and tip/base keypoints. This is valuable because synthetic data is most useful when it carries dense ground truth that is expensive or impossible to annotate at scale in real fluoroscopy [22, 23, 24].

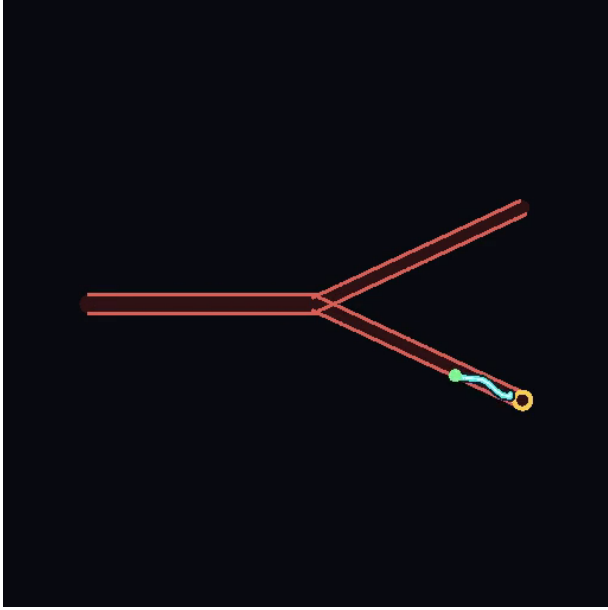
3.3 Data workflow

The CLI is intended to make the first research loop short: `lumen play` renders a rollout; `lumen benchmark` writes scorecards; `lumen capture` writes replayable case bundles; `lumen validate` checks assets, observations, labels, and sidecars; `lumen index` writes a JSONL dataloader index; `lumen split-index` produces episode-grouped train/validation /test splits; and `lumen materialize-batch` writes strict NPZ smoke-test batches. This workflow turns simulation into a dataset interface rather than a demo script.

4 Benchmark semantics

The canonical benchmark separates goal achievement from safe goal achievement. Each task reports:

- **success rate:** the tip reached the target;
- **safe success rate:** the target was reached without crossing the wall-safety threshold;



(a) Schematic rollout



(b) Synthetic fluoroscopy

Figure 3: **Paired views of the same control problem.** The schematic view makes rollout state interpretable; the fluoroscopic view is the image stream used by CV and image-observation policies.

- **unsafe success rate:** the target was reached only with a safety breach;
- **maximum penetration:** worst wall-safety violation observed;
- **return:** task reward after safety and efficiency terms.

The leaderboard ranks safe success first, then raw success, then lower maximum penetration, then return. This is a deliberately clinical ordering: an agent that arrives by perforating the wall has not solved the problem.

5 Positioning relative to CathSim and prior simulators

CathSim is the strongest open reference point for endovascular RL environments: it packages a MuJoCo/dm_control catheter environment, force sensing, and baseline agents for autonomous catheterization research [1, 2]. Lumen builds on the research direction CathSim helped establish, while targeting a broader substrate for deformable mechanics and multimodal data generation.

Prior work in autonomous endovascular navigation has explored robotic catheterization, reinforcement learning, image-guided control, and simulation-to-real transfer [3, 4, 7, 8, 9]. Lumen’s contribution is to expose a compact open stack where those experiments can share the same mechanics, sensor outputs, and data contracts.

6 Implementation surface

The public repository provides:

- procedural straight, stenotic, tortuous, and branching assets;
- Newton/Warp-backed guidewire simulation;

Table 1: **Research surface comparison.** The point is not API parity; it is the simulation and data surface available to an RL/CV researcher.

Capability	CathSim-style open RL envs	Lumen
Physics emphasis	Rigid or mesh-centric catheter task	Deformable lumen field plus tube-intrinsic contact
Solver substrate	MuJoCo/dm_control family	NVIDIA Newton/Warp differentiable simulation
Benchmark semantics	Goal/reward and force signals	Safe success, unsafe success, penetration, return
Vision data	Pixel/segmentation surfaces in simulator	Synthetic fluoro, masks, keypoints, node positions, case bundles
Dataset workflow	Environment-centric	Capture, validate, index, split, materialize
License posture	Project-dependent	Apache-2.0 open core

- Gymnasium navigation environments;
- synthetic fluoroscopy and luminal cameras;
- replayable case bundles with masks/keypoints;
- benchmark scorecards and leaderboards;
- validation and dataloader-index CLIs;
- typed Python package metadata and an Apache-2.0 license.

The quick-start installation path is:

```
git clone https://github.com/SeldingerMed/seldinger-lumen
cd seldinger-lumen
pip install -e ".[dev]"
lumen doctor
lumen play stenotic --out lumen-run
lumen benchmark lumen-bench
```

7 Conclusion

Lumen is a launch point for open endovascular AI research: a differentiable, GPU-parallel simulator that treats wall mechanics, contact, image formation, labels, and safety-scored benchmarks as one integrated problem. It is designed for researchers who want to train agents, generate CV data, test policies, and compare methods without starting from a rigid-pipe toy task or a closed simulation stack. The open repository and demos are available at <https://github.com/SeldingerMed/seldinger-lumen> and <https://seldingermed.github.io/seldinger-lumen/>.

Acknowledgments

Lumen builds on open scientific software and published work in differentiable simulation, catheter robotics, arterial biomechanics, synthetic data, and reinforcement learning. The project uses NVIDIA Newton/Warp as its solver substrate and releases the open core under Apache-2.0.

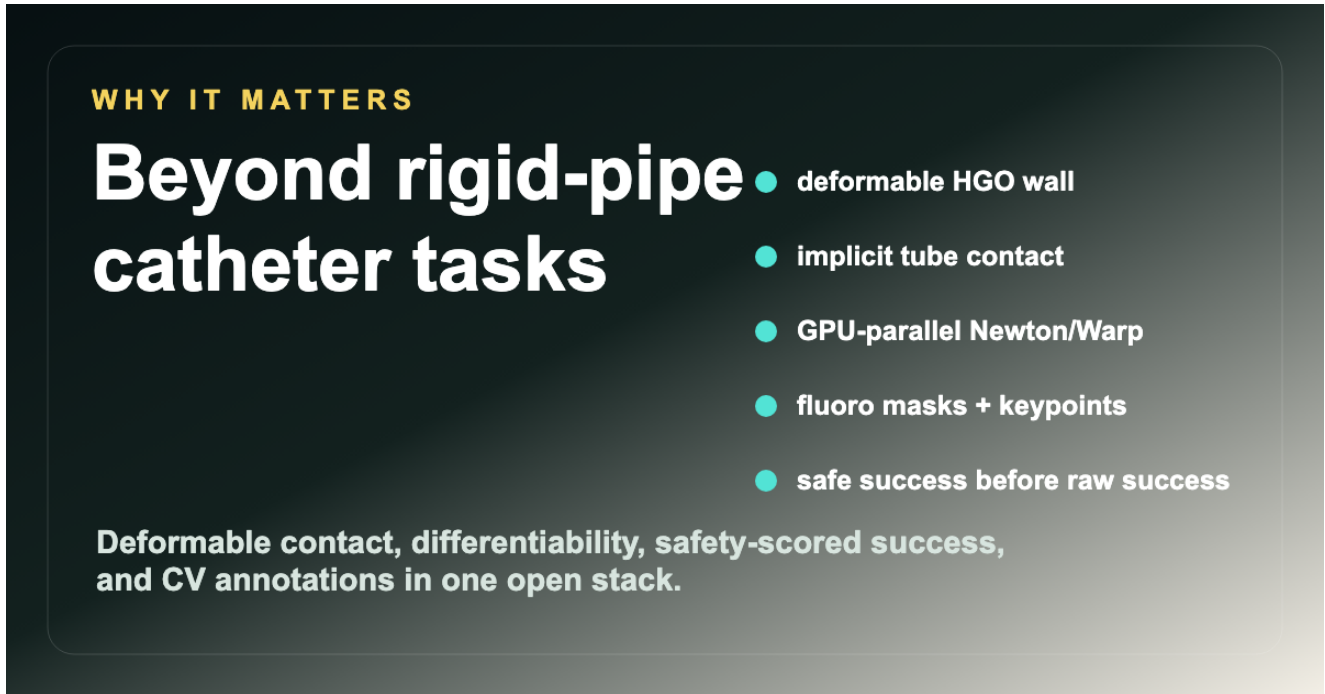


Figure 4: **What Lumen adds to the open endovascular simulation surface.** Deformable-wall contact, differentiability, safety-scored success, and CV labels are co-designed rather than bolted together.

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